



Department of Electrical & Computer Engineering
Rajshahi University of Engineering & Technology

Course Plan / Outline

1. **Course Information:**

Course Title	Software Engineering & Information System Design Sessional
Course Code	ECE 3118
Course Type	Program Course (Sessional)
Credit	0.75 (Three Contact Hours Per Two Weeks), three lecture slots per two weeks according to class routine.
Semester	3 rd Year Odd Semester

2. **Instructor Information:**

Name of the Instructor	To be announced before starting of the semester
Designation	
Office Room	
Cell	
Email	
Webpage link	

3. **Prerequisite (if have any): None**

4. **Course Rationale:**

Development of software in a systematic way most often requires effective team work and preservation of different versions. This course provides essential knowledge about different version control systems (VCS) like git and how to work efficiently in a team. It also aids in the development of effective communication skills in order to discover a client's specific demands.

5. **Course Objectives:**

After successfully completing this course, students will-

1. Understand the functionalities of the version control system GIT.
2. Make proper documentation for efficient knowledge transferring.
3. Be able to use online repository systems to collaborate a remote team.

6. **Course Content / Course Description:**

Sessional based on the theory of course ECE 3117. Course content of ECE 3117 is as follows:

Introduction: Introduction to Software and its Nature, Software Engineering Methods, Different Types of Software Process Model.

Software Requirement Analysis and design: Software Requirements Analysis and their Applications, Software Prototyping.

Design of Software: Software Design and its Different Techniques, Software Configuration Managements. System Structuring, Control Models

Software Testing: Verification and Validation Planning, Software's Testing Strategies and Different Type of Testing Techniques,

Software Quality Assurance: Management and its Quality Assurance, Concepts of Software Reengineering.

Information System Design:

Application Development Policy and Strategies: Planning of Information System, Policy in Information System Development, Strategies for Achieving Information System Goals.

Application System Development Life Cycle: Phases in Application System Development, Interrelationship among Each Phase. Feasibility Assessment: Problems and Needs in Information System Development.

Feasibility Assessment: Economic, Technical, Operational and Schedule Feasibility.

Information Requirements Determination: Strategies for Obtaining Information Requirements, Methods for Providing Assurance that Requirement are Correct and Complete.

Structured Systems Analysis & Design: Steps in Structured Systems Analysis, Activity Diagrams and Related Documentation, Problem Analysis, New System Design.

7. **Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)**

CLO No.	Course Learning Outcomes (CLO)	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Verify the knowledge of version control and online repository to collaborate with an individual or a team.									A5			
CLO2	Pursue the pit-falls of an information system through communicating with the corresponding officials and staffs.										A3		

8. **CLOs Assessment Policy:**

CLOs	Lab performance	Lab report	Lab final	Quiz	Presentation	Viva
CLO1	√	√		√		√
CLO2		√				√

9. **References:**

SL No.	Text Book/Reference Book(s)	Author(s)	Online Availability
1.	Systems Analysis and Design	Elias M. Awad	Link is provided in Google Class Room

10. **Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning assessment strategy mapped with CLOs**

Week	Topic	Delivery Material	Teaching Learning Strategy	Assessment Strategy	Aligned CLOs
1-2	Introduction to distributed version control system (DVCS)	Lab Manual	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project	☒ Quiz ☒ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration	CLO1

			☐ Audio/Video ☐ Others	☐ Lab test ☐ Others	
3-4	Documentation writing using markdown language.	Lab Manual	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☒ Quiz ☒ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☐ Lab test ☐ Others	CLO1
5-6	Basic git commands – Init, Add, Pull, Push, Clone, Remote, Branch, Status, Log, Fetch etc.	Lab Manual	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☒ Quiz ☒ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☐ Lab test ☐ Others	CLO1
7-8	Online repository (Github) and git branching.	Lab Manual	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☒ Quiz ☒ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☐ Lab test ☐ Others	CLO1
9-10	Applying different git commands to achieve specific objectives.	Lab Manual	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☐ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☒ Quiz ☒ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☐ Lab test ☐ Others	CLO1
11-12	Group report on information system design.		-----	☐ Quiz ☒ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☐ Lab test ☐ Others	CLO2
13-14	Quiz		-----	-----	-----

11. Virtual Availability of Lecture Materials:

All resources related to this course are available in online platform.

Medium	Google classroom/Zoom/MS Teams etc.
Instructions	
Class Code	

12. **Assessment & Marks Distribution:**

SL No.	Name of Assessment	Marks	Date
1.	Class Participation and Attendance	10	As Per Class Routine
2.	Quiz	20	As Per Examination Routine
3.	Lab Performance, Lab Report, Lab Final, Presentation, Viva	45	On Lab Days
4.	Board Viva	25	As Per Examination Routine
Total Marks:		100	

13. **Grading Policy:**

Mark wise grading policy of this course is as follows:

% of Marks	Letter Grade	Point
80 – 100	A+	4.00
75 – 79	A	3.75
70 – 74	A-	3.50
65 – 69	B+	3.25
60 – 64	B	3.00
55 – 59	B-	2.75
50 – 54	C+	2.50
45 – 49	C	2.25
40 – 44	D	2.00
0 – 39	F	0.00

14. **Additional Course Policy:**

SL. No.	Name of the Test	Description
1.	Class Participation and Attendance	Attending classes are mandatory for the students. Students will be marked in a scale of 10 marks according to Academic Ordinance for the Undergraduate Studies.
2.	Quiz	For this course quiz mark is 20. The students need to seat for an examination with the syllabus what was taught throughout the lab in this course.
3.	Lab Performance, Lab Report, Lab Final, Presentation, Viva	All the assessment strategies are considered as regular assessment. The course teacher will assign works to the students and evaluate them with a scale 45 marks that will be counted as regular assessment marks in final scoring.
4.	Board Viva	It is mandatory for individual student to attend the board viva. The students will be evaluated in a scale of 25 marks. Absence in the board viva will result in “F” grade in all sessional courses of the corresponding semester.

15. **Students Policy with Physically Handicapped:**

Students with disabilities are required to inform the Department of Electrical & Computer Engineering of any specific requirement for classes or examination as soon as possible.

16. **Generic Skills (Indirect):**

SL. No.	Generic skills	Attained Generic Skills (Please tick against the relevant skill)	How the skills will be developed
1.	Presentation skill		
2.	Writing skill	√	Report Writing
3.	Interpersonal skill	√	Information gathering for report

4.	Leadership skill	√	Team management for report
5.	Community skills	√	Suggesting improvement scope for information system of the community.

17. **Quick Links:**

- Website: <https://www.ece.ruet.ac.bd>